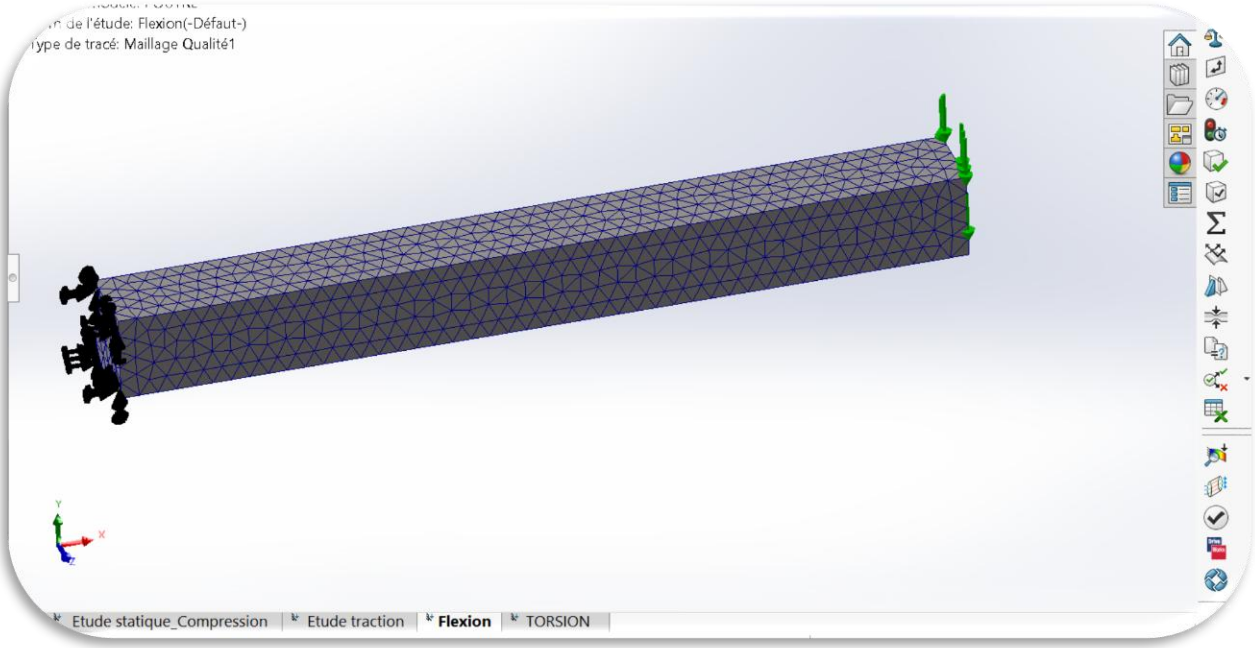
Program : Mechanical Design & Simulation Internship — CodeAlpha

-Task 2 : Structural Simulation (FEA)
-Software : SolidWorks Simulation



Period : June 10 – July 10

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CODE ALPHA

CODE ALPHA — ENGINEERING INTERNSHIP

FEA Report – Structural Analysis of a Rectangular Beam

Task 2 — Structural Simulation by Finite Element Analysis (FEA)

1. Project Objective

The objective of this study is to assess, using the Finite Element Analysis (FEA) method, the mechanical behavior of a rectangular beam subjected to four types of static loading: compression, tension, bending, and torsion. The analysis aims to identify the stress levels, the induced deformations, and the factor of safety associated with each case, in order to determine the critical configurations of the component.

2. Model Under Study

The analyzed part is a solid beam of simple geometry, fixed at one end and loaded at the opposite end. The characteristics of the model are summarized below:

Length	100 mm
Cross-section	10 × 10 mm
Material	Steel (Re ≈ 206 MPa)
Boundary conditions	Fixed support at one end, load applied at the other

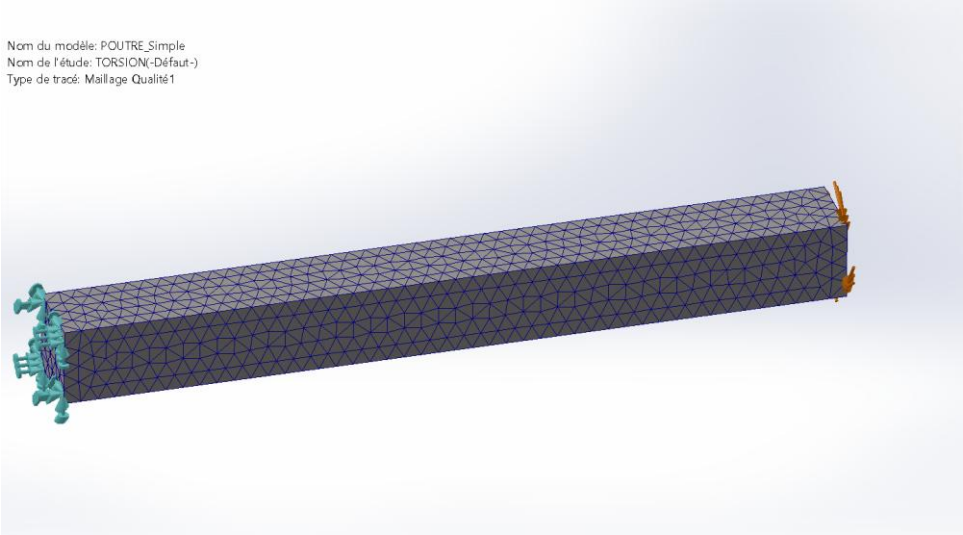


Fig. 1 — Mesh of the beam used for the finite element simulation

2.1 Boundary Conditions and Applied Loads

The boundary conditions were defined identically for all studies, with only the nature of the loading varying according to the tested case. The selected configuration reproduces a cantilever beam, representative of a rigid fixation in service.

- Fixed support (fixed geometry — prescribed displacement): full restraint of all 6 degrees of freedom (3 translations, 3 rotations) on the end face, simulating a perfectly rigid clamped support.
- Axial loading (compression / tension): a 1,000 N point force applied on the free face, oriented along the longitudinal axis of the beam (direction reversed between tension and compression).
- Bending load: a force of the same magnitude applied perpendicular to the neutral axis, generating a maximum bending moment at the fixed support.
- Torsional load: a 100 N·m torque applied about the longitudinal axis of the beam on the free face.

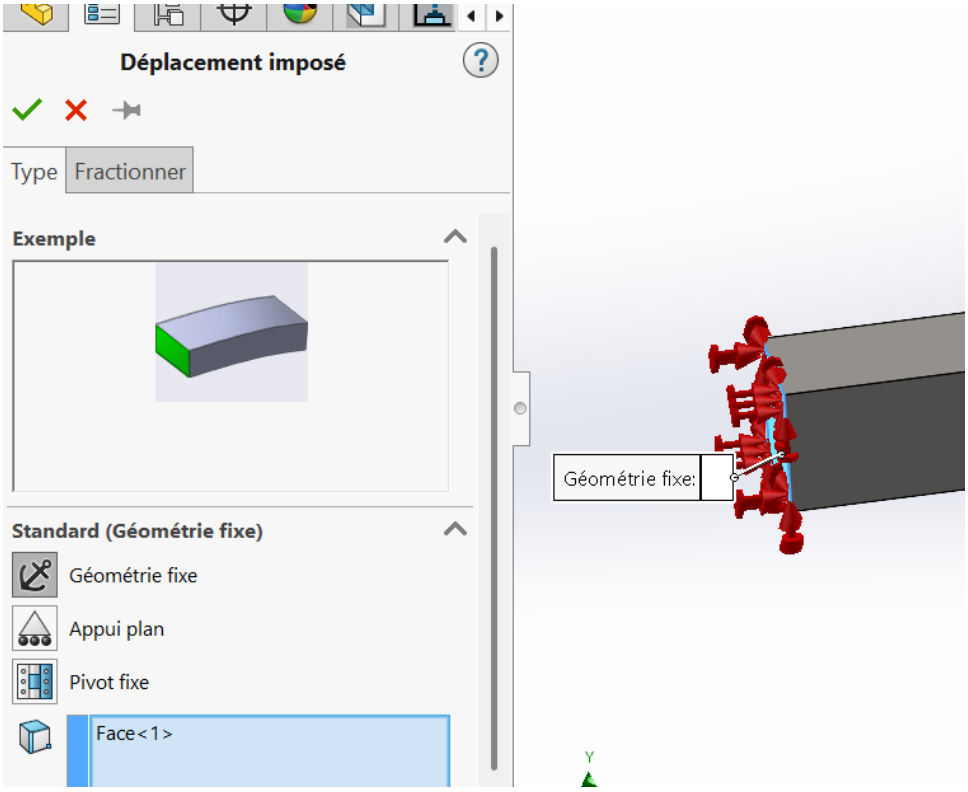


Fig. 2 — Definition of the fixed support (fixed geometry, prescribed displacement) on the end face

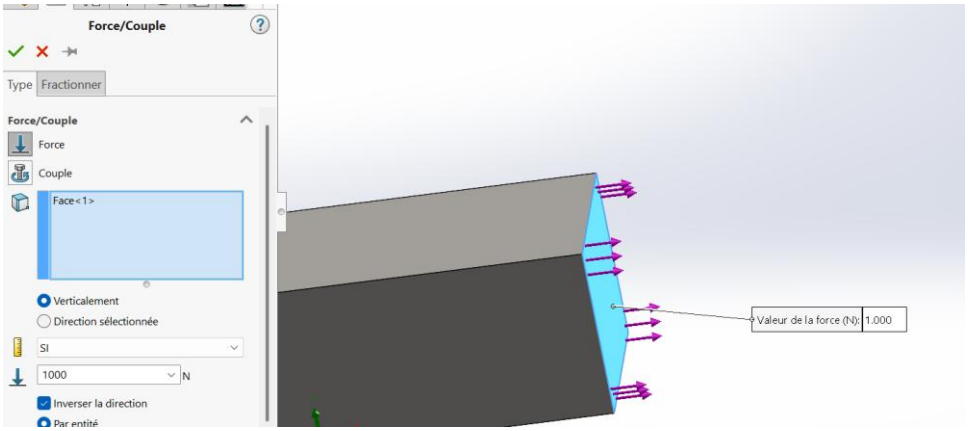


Fig. 3 — Application of the axial load (1,000 N force) on the free face

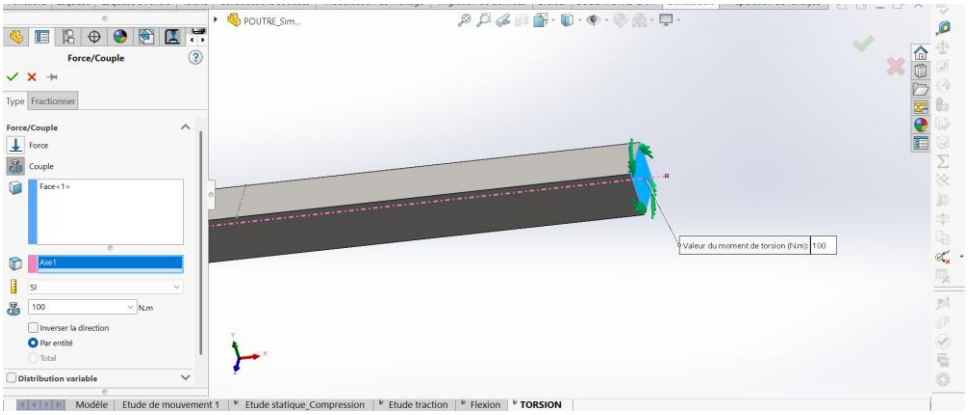


Fig. 4 — Application of the torsional torque (100 N·m) about the beam axis

3. Simulation Results

Each loading case was simulated independently in order to isolate its effect on the stress and strain distribution.

3.1 Compression

- Maximum stress: ~14 MPa (well below Re = 206 MPa)
- Strain / displacement: virtually zero
- Factor of safety: ≈ 14

→ **Conclusion:** perfectly elastic behavior, no critical zone detected.

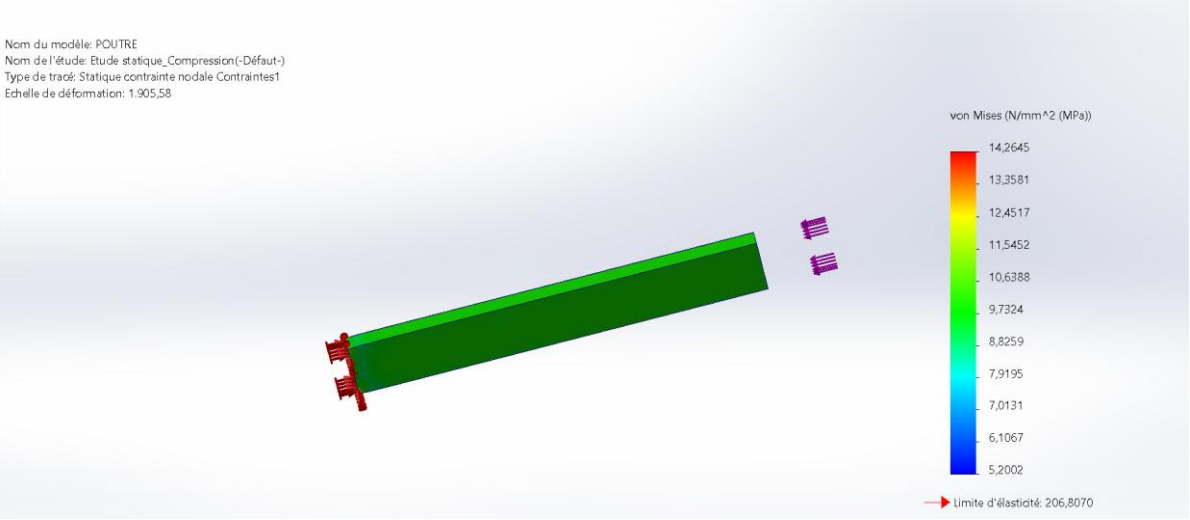


Fig. 5 — Von Mises stresses under compression

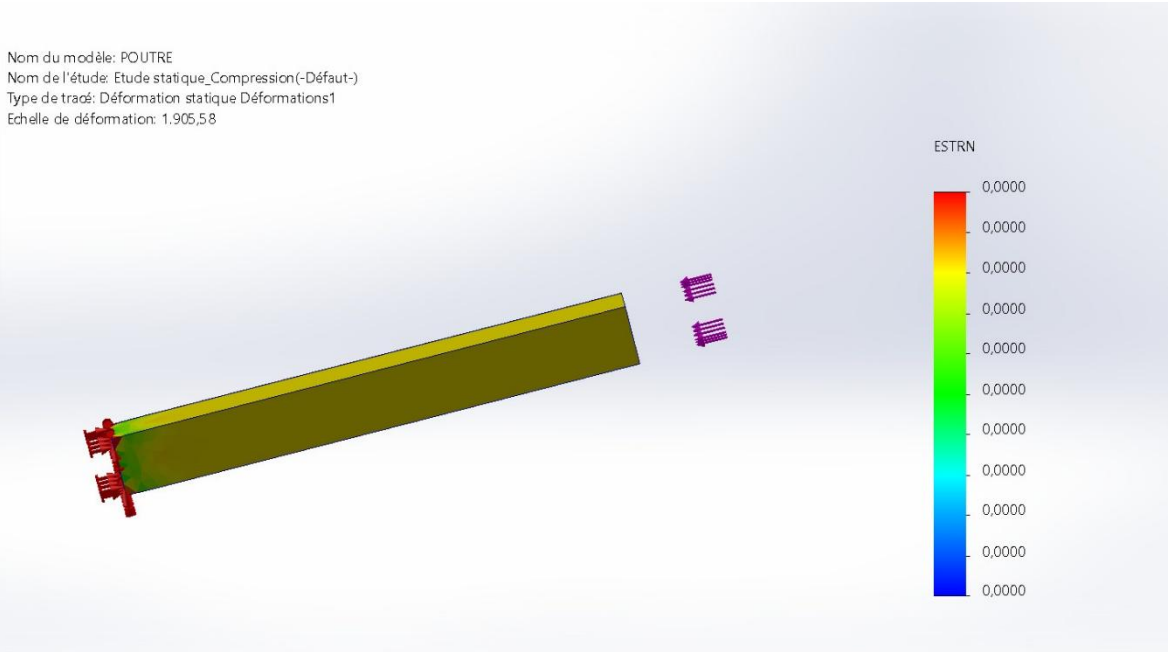


Fig. 6 — Strain under compression

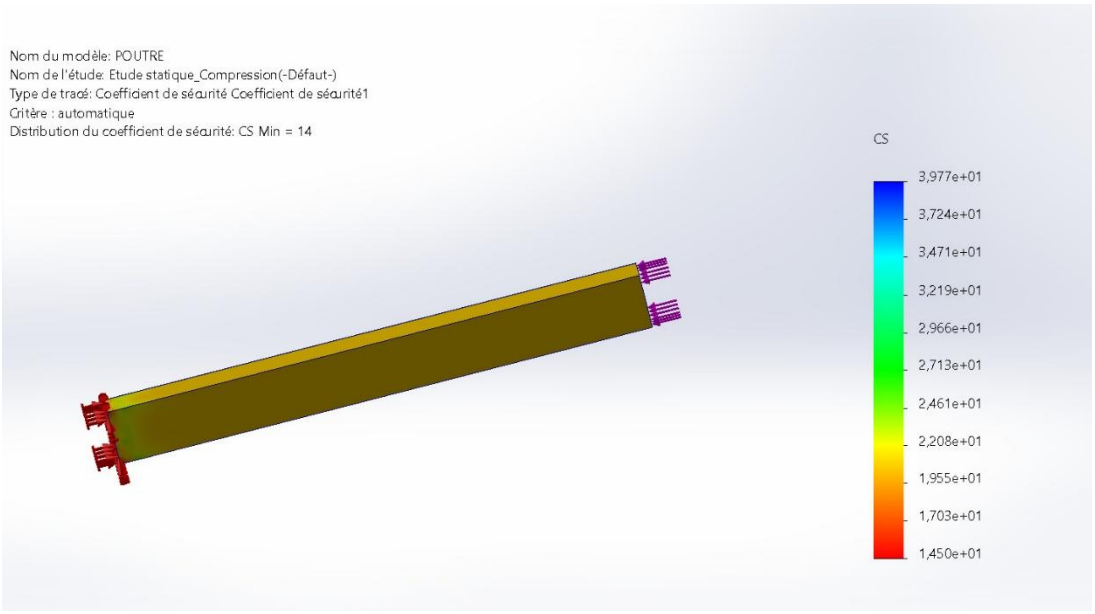


Fig. 7 — Factor of safety under compression (min FoS = 14)

3.2 Tension

- Maximum stress: ~14 MPa (identical to compression)
- Strain / displacement: ~0.005 mm (maximum displacement)
- Factor of safety: ≈ 14

→ **Conclusion:** the beam withstands the load without difficulty, with a comfortable safety margin.

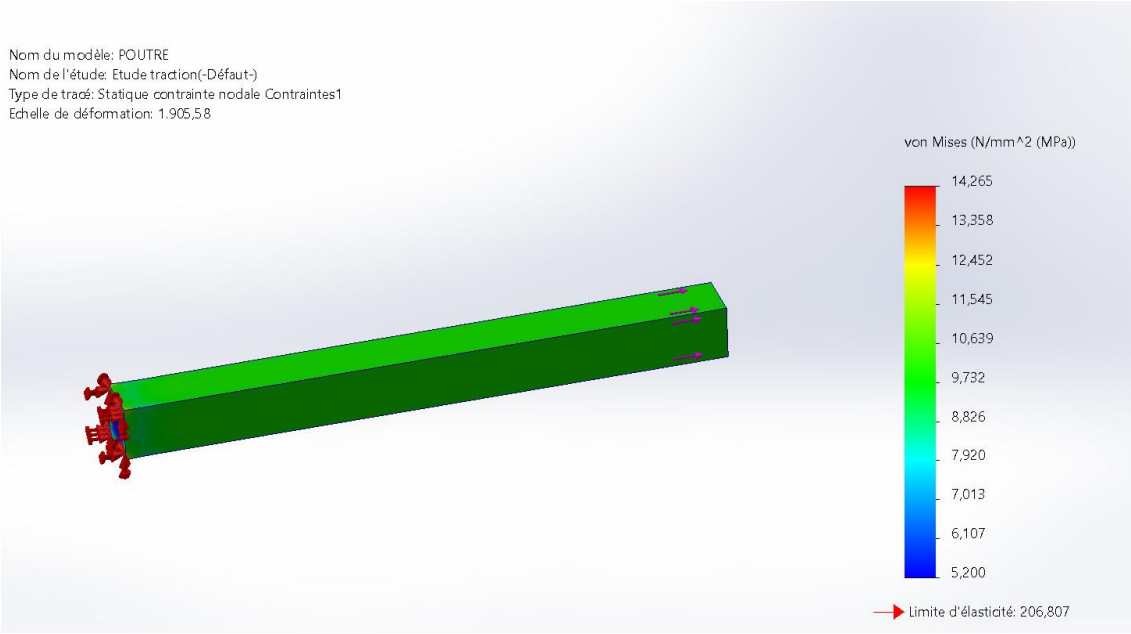


Fig. 8 — Von Mises stresses under tension

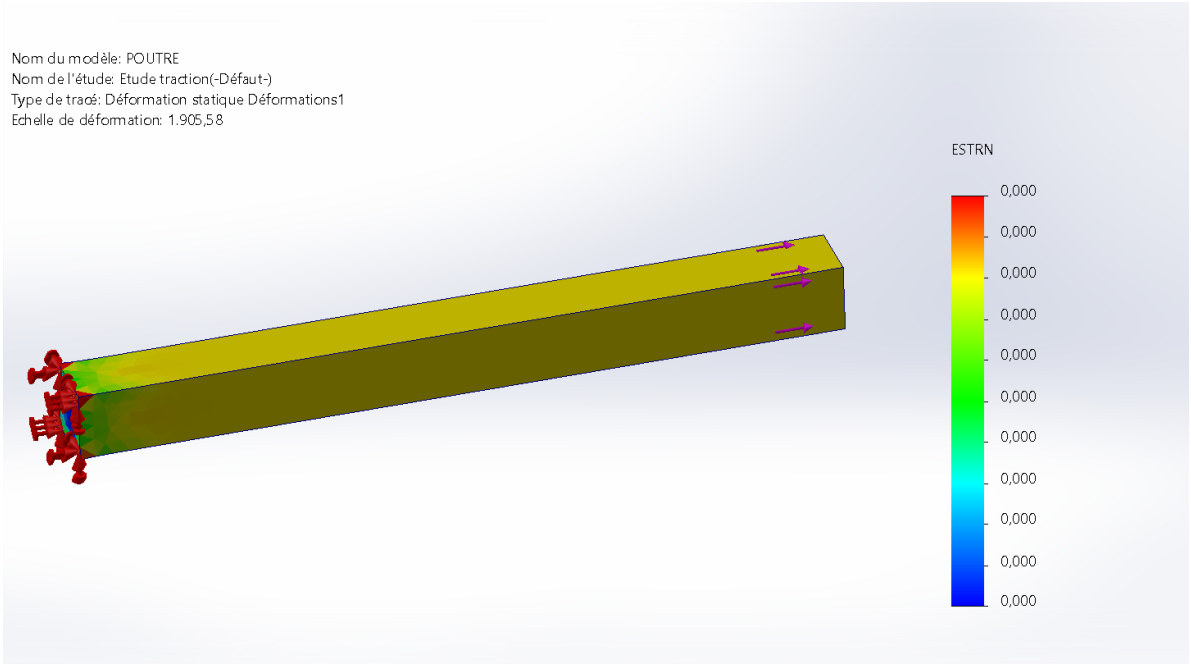


Fig. 9 — Strain under tension

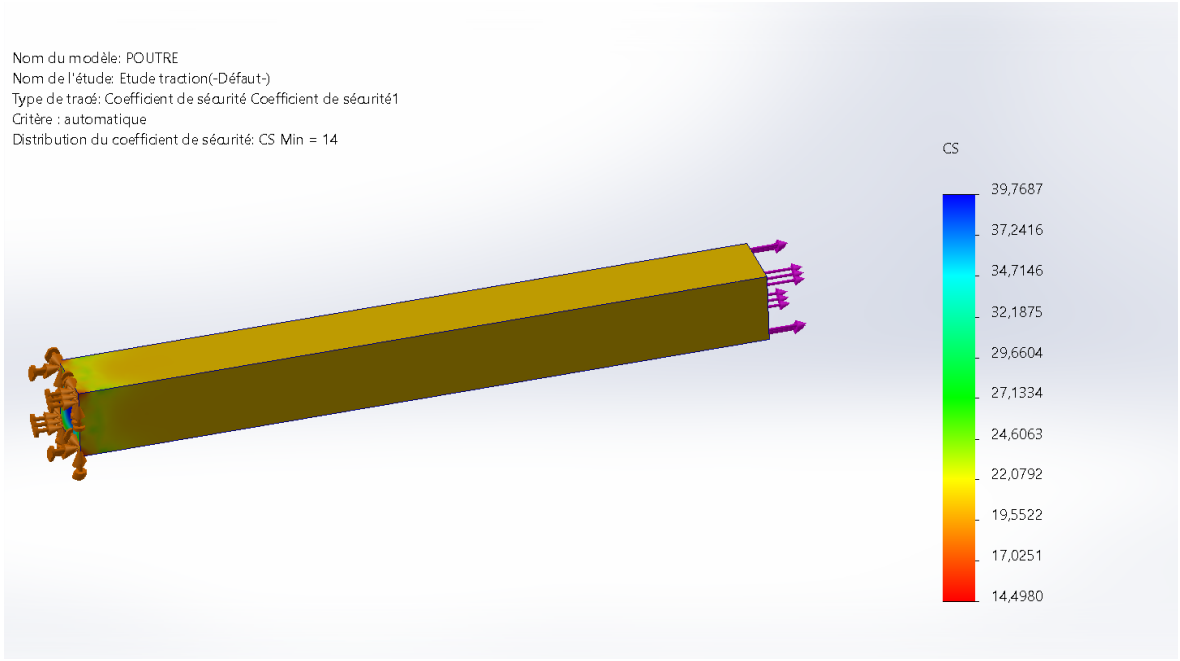


Fig. 10 — Factor of safety under tension (min FoS = 14)

3.3 Bending

- Maximum stress: ~630 MPa (above Re = 206 MPa)
- Strain / displacement: ~2.1 mm
- Factor of safety: ≈ 0.33

→ **Conclusion:** failure likely; critical zone located at the fixed support.

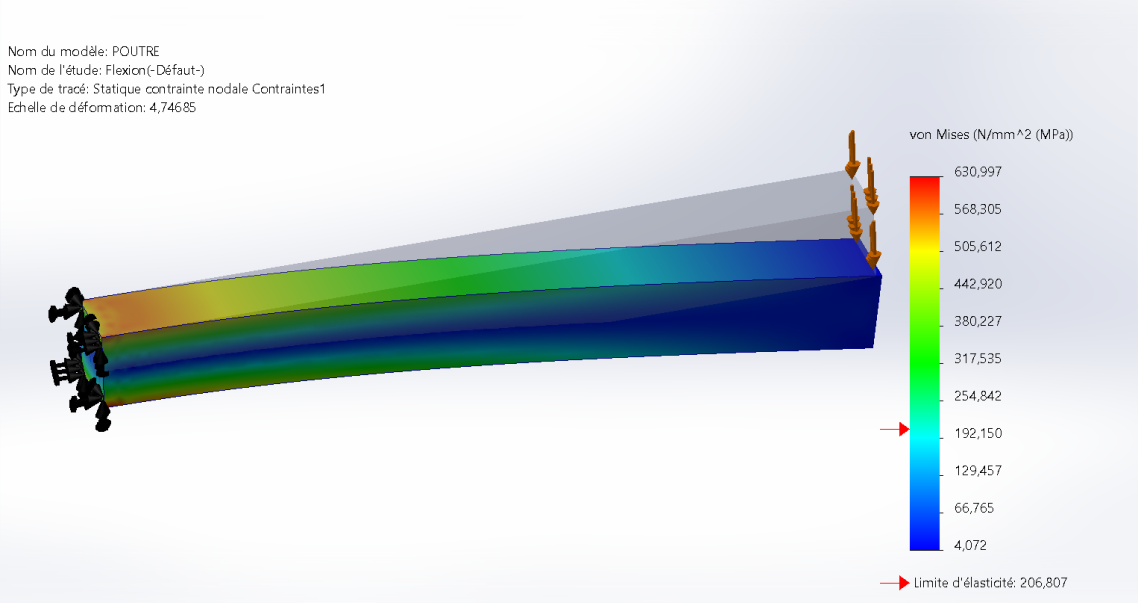


Fig. 11 — Von Mises stresses under bending (fixed-support zone)

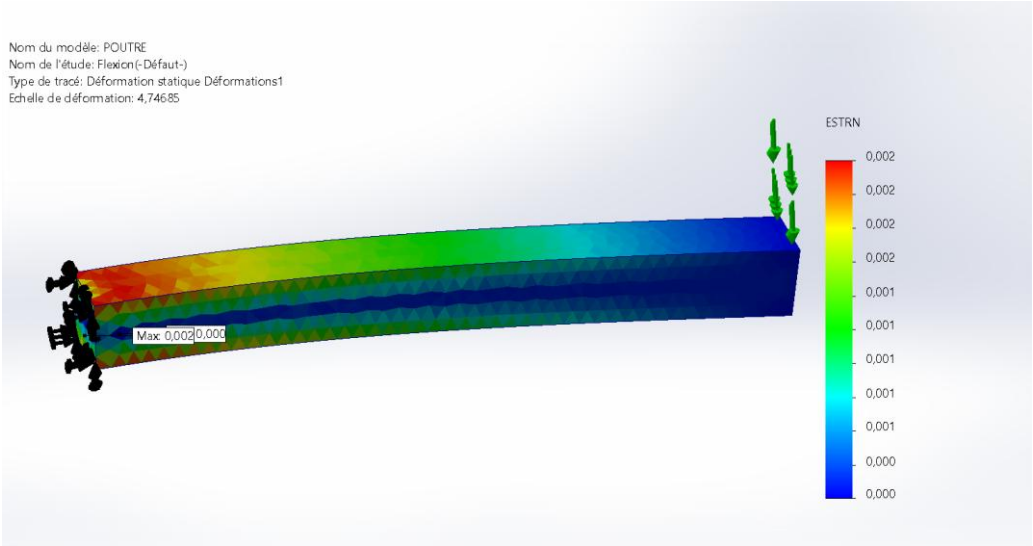


Fig. 12 — Strain under bending

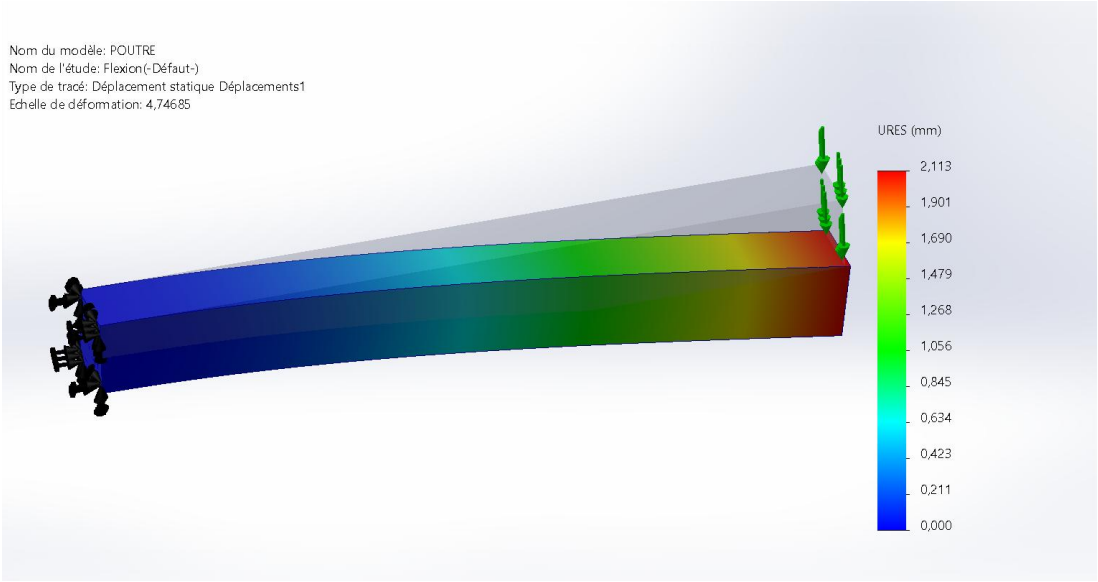


Fig. 13 — Resultant displacement under bending (max ≈ 2.11 mm)

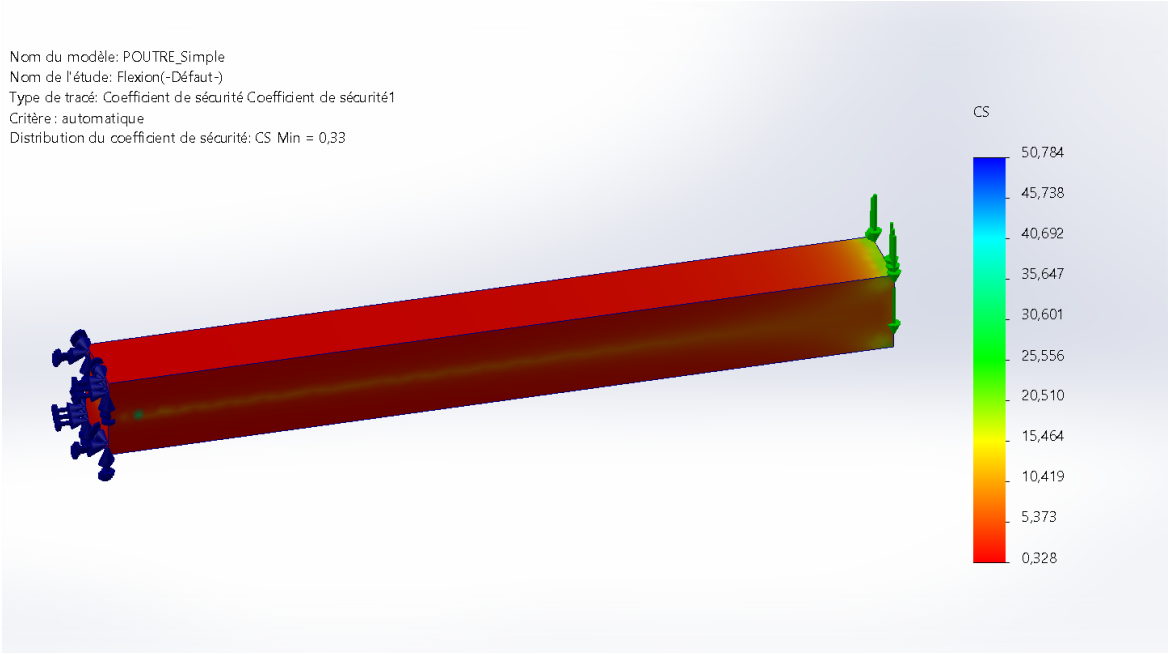


Fig. 14 — Factor of safety under bending (min FoS = 0.33)

3.4 Torsion

- Maximum stress: ~896 MPa (far above Re = 206 MPa)
- Strain / displacement: ~0.76 mm
- Factor of safety: ≈ 0.23

→ **Conclusion:** most severe loading case; mechanical failure expected.

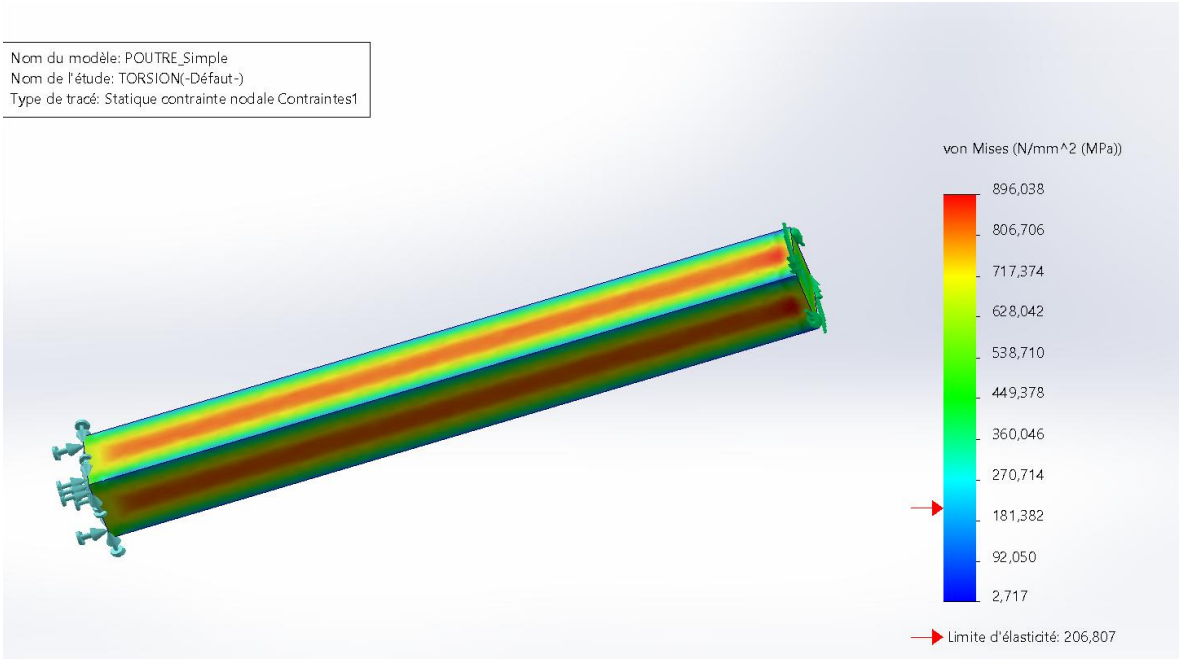


Fig. 15 — Von Mises stresses under torsion

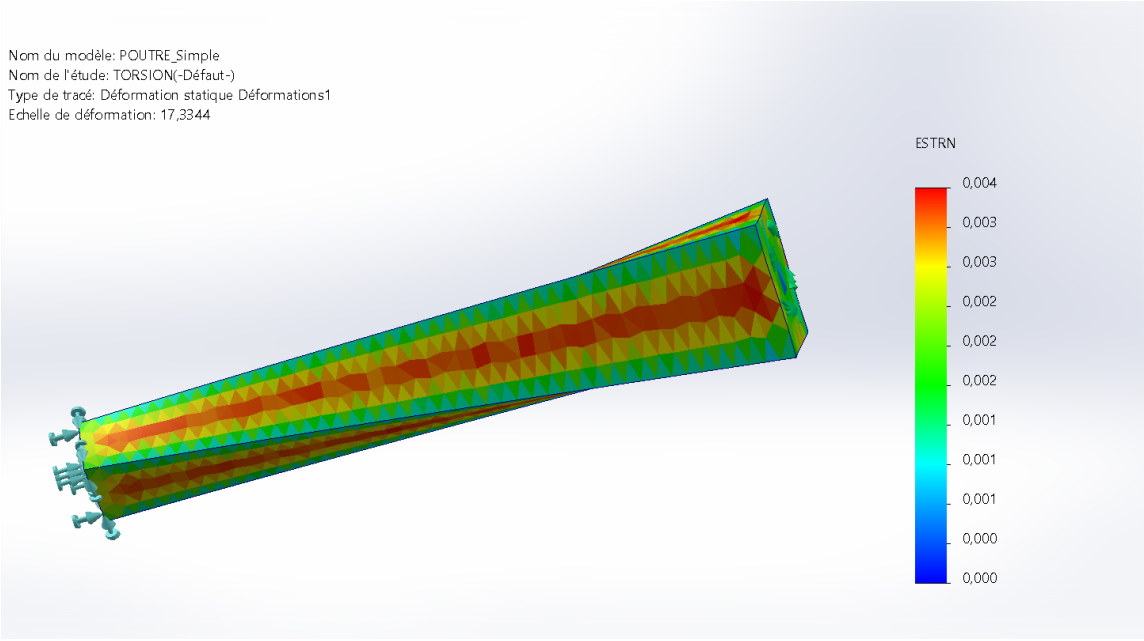


Fig. 16 — Strain under torsion

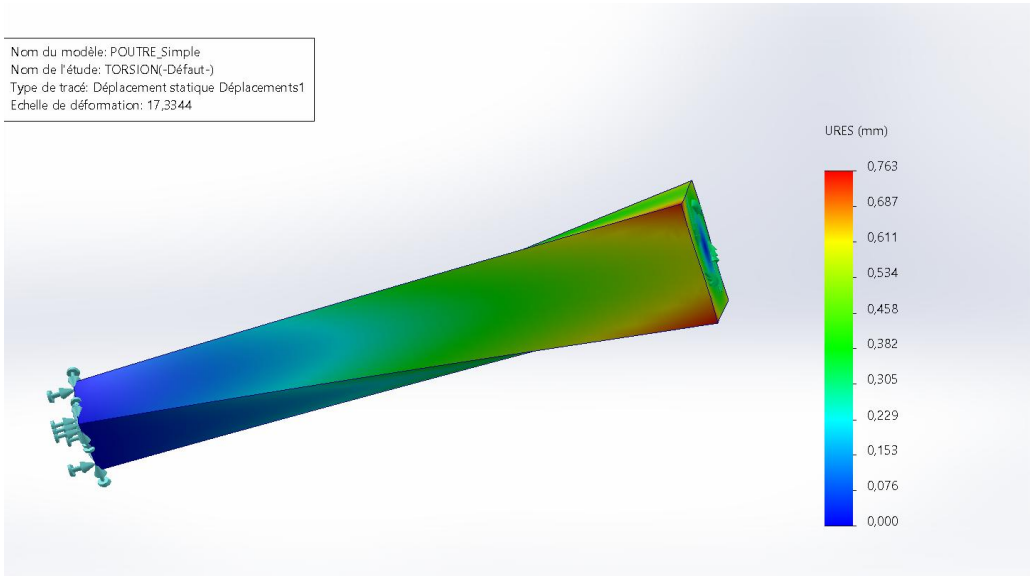


Fig. 17 — Resultant displacement under torsion (max ≈ 0.76 mm)

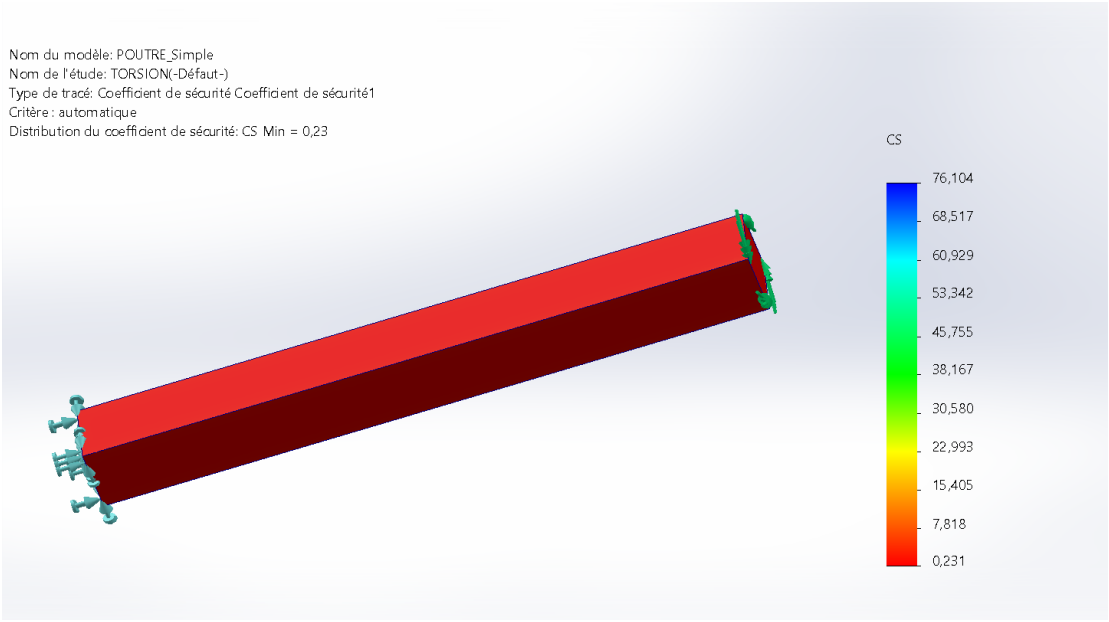


Fig. 18 — Factor of safety under torsion (min FoS = 0.23)

3.5 Comparative Summary

Loading	Max. stress	Strain / Displacement	Factor of safety (FoS)	Verdict
Compression	~14 MPa	≈ 0 mm	**≈ 14**	**Safe**
Tension	~14 MPa	~0.005 mm	**≈ 14**	**Safe**
Bending	~630 MPa	~2.1 mm	**≈ 0.33**	**Failure likely**
Torsion	~896 MPa	~0.76 mm	**≈ 0.23**	**Failure expected**

4. Conclusion

The beam exhibits safe and well-controlled behavior under compression and tension loading, with high factors of safety (FoS ≈ 14) and negligible deformations. However, it fails under bending (FoS ≈ 0.33) and torsion (FoS ≈ 0.23) loading, with stresses far exceeding the yield strength of the material (Re ≈ 206 MPa).

In both critical cases, the most highly stressed zone is systematically located at the fixed support, where the bending moments and torsional stresses are at their maximum. These results highlight the need for geometric optimization (increased cross-section, addition of fillets) or the selection of a material with a higher yield strength, particularly for configurations subjected to bending and torsion.